Comments: ① Check notes: right D-module structure on ω_X

left structure:

$(f\Theta)\cdot m = f(\Theta\cdot m)$

$\mathcal{L}_\Theta(\omega) = d\, i_\Theta \omega + i_\Theta d\omega$

$\mathcal{L}_\Theta(x) = i_\Theta d\omega$

$\mathcal{O}_X$ Lie derivative ω_X

right structure

$(f\Theta)\cdot m = \Theta\cdot(f\cdot m)$

Exercise

$\mathcal{L}_\Theta(\omega) = d\, i_\Theta \omega$

② Explanation for right action of Θ on D_f "transfer bimodule"

$D_f = \mathcal{O}_X \otimes_{\mathcal{O}_Y} D_Y$

$\Theta\cdot(f\otimes D) = \Theta(f)\otimes D + f\cdot\tilde{\Theta} D$

$\Theta f\otimes D = \Theta(f)\otimes D + f\Theta\otimes D$

$= \Theta(f)\otimes D + f\cdot\tilde{\Theta} D$

$\Theta\otimes D$: Θ acting on D_Y

③ $j: \mathbb{G}_m \hookrightarrow \mathbb{A}^1$ $\quad j_\bullet \omega_{\mathbb{G}_m} = j_* \mathcal{O}_{\mathbb{G}_m}$ $\quad \ni \frac{1}{x}$

D_x acts thru restriction to D_m.

Want to present $j_\bullet \omega_{\mathbb{G}_m}$ ... look for generators get relations

[no shift]

$\rightsquigarrow$ relation $\frac{1}{x}\cdot x\partial = 0$, $\quad j_\bullet\omega_{\mathbb{G}_m} \simeq \frac{D_x}{x\partial\cdot D_x} \xrightarrow{\text{left}} \frac{D_x}{D_x\cdot\partial x} = \mathcal{O}_{\mathbb{G}_m}$

Kashiwara:

$DMod_\Lambda(X) = $ D-mod $\mathcal{M}$ w/ $SS(\mathcal{M})\subseteq\Lambda$ $\qquad \Lambda\subseteq T^*X$

never really run into issues:

$DMod_Z(X) = $ D-mod $\mathcal{M}$ w/ classical $\mathrm{supp}(\mathcal{M})\subseteq Z$ $\qquad Z\subseteq X$

later:

$\dim(SS(\mathcal{M})) \geq \dim(X) = \frac{1}{2}\dim(T^*X)$

Ex $SS(D_X) = T^*X$

Def: holonomic SS is Lagrangian

Def: $i: Z\hookrightarrow X$

$i_!: DMod(Z)\to DMod(X)$ $\quad$ defined to be $i_! = i_\bullet$

$i^! = \mathcal{H}om_{i^{-1}D_X}(D_i, i^{-1}\mathcal{I})$ $\leftarrow$ Claim: $(i_!, i^!)$ adjoint.

Define D-module structure...

Prop: (1) $(i_!, i^!)$ adjoint functors $\Rightarrow$ $i_!$ right exact, $i^!$ left exact

Exercise

(2) $i_!$ is exact.

Proof of (2) $i_\bullet(\mathcal{M}) = i_*\big(\underbrace{\mathcal{M} \otimes_{D_Z} \mathcal{O}_Z \otimes_{i^{-1}\mathcal{O}_X} i^{-1}D_X}\big)$ ✓ $\boxed{\text{exact}}$

Check exactness local rings:

fix $x \in Z \subseteq X$

$$M \otimes_{D_{Z,x}} \frac{\mathcal{O}_{Z,x}}{I} \otimes_{\mathcal{O}_{X,x}} D_{X,x}$$

Fact: $Z \hookrightarrow X$ l.c.i.

local rings: $\mathcal{O}_{Z,x} = \mathcal{O}_{X,x}/\underbrace{(f_1, \dots, f_r)}_{I}$

where $r = \operatorname{codim}(Z \subseteq X)$

Nakayama: get generators $m_{X,x} \ni x_1, \dots, x_n$

such that $f_1 = x_1 \dots f_r = x_r$

$T_{Z,x} \subseteq T_{X,x}$

$$= M \otimes_{D_{Z,x}} I \cdot D_{X,x} \backslash D_{X,x}$$

has symbols $\partial_1, \dots, \partial_n$

For exactness: claim $I \cdot D_{X,x} \backslash D_{X,x}$ is free as left $D_{Z,x}$-module with basis $\partial_1, \dots, \partial_r$

write down a map respecting order filtration. $\rightsquigarrow$ gr

Ex. $\{0\} \hookrightarrow \mathbb{A}^1$

$k \mapsto D/Dx \cong k[z][\partial]$

Exercise: check stuff

Exercise: Show that if j is a this open embedding, then $j_\bullet$ is also exact.

$\boxed{i \text{ closed immersion}}$

Thm (Kashiwara) $i_!: DMod(Z) \xrightarrow{\simeq} DMod_Z(X)$ equiv. of categories

Remark: False for QCoh.

$i_*: QCoh(Z) \longrightarrow QCoh_Z(X)$ NOT equiv.

$\{0\} \to \mathbb{A}^1$

vector spaces $\longmapsto k[x]/x^n$

Proof: (1) Pass to local rings, go to left modules (i.e. choose sections

$i_! = D_X/D_X \cdot I \otimes_{D_Z} M$

$i^! = \mathcal{O}_X/I \otimes_{\mathcal{O}_Z} M$

$I = (x_1, \dots, x_r)$

(x)

$\omega_X \otimes D_X \otimes \omega_X^{-1} \xleftrightarrow{\simeq} D_X^{op}$

choose dual sections;

$D_X \xrightarrow{\simeq} D_X^{op}$

$1 \longmapsto 1$

$X = \mathbb{A}^n$

② reduce to claim 1, i.e. locally $Z \hookrightarrow Z_1 \hookrightarrow Z_2 \hookrightarrow \dots X$

smooth!

Idea: $D = \partial x \curvearrowright M$

$D^t = x\partial$

$x\partial = \partial x - 1$

Suppose m is λ-eigenvector for D.

- m is an $\lambda - 1$ eigenvect. for D^t
- xm is a $\lambda + 1$ eig. for D
- ∂m is $\lambda - 1$ eigen for D

"raising operator"

$\partial = \frac{d}{dx}$ lowering operator

$x\partial m = (\partial x - 1)\cdot m = (\lambda - 1)m$

$\partial x \cdot xm = (x\partial + 1)\cdot xm = x(\partial x m) + xm = \lambda x m + xm = (\lambda+1)m$

$$\begin{tikzcd} \cdots & \partial m \arrow[l, bend left] \arrow[r, bend left, "x"] & m \arrow[l, bend left, "\partial"] \arrow[r, bend left, "x"] \arrow[loop above, "D=\lambda"] \arrow[loop below, "D^t = \lambda - 1"] & xm \arrow[l, bend left, "\partial"] & \cdots \end{tikzcd}$$

All of the $x^n m$, $\partial^n m$ nonzero except when forced to be by eigenvector 0.

Claim: Every $m \in M$ is decomposable into strict eigenvectors

But: x acts by fixing m in M $\rightsquigarrow$

$x^k m = 0$ (k min.)

$\partial^{k-1} x^{k-1} = x^{k-1}\partial^{k-1} + (k-1)!$

Claim: $x^{k-1}\cdot m$ is nonzero eigenvector w/ eig. 0

$\partial x (x^{k-1} m) = 0$

$\Rightarrow \partial^{k-1} x^{k-1} m$ eigenvalue $-k+1$

$0 = x_0$

$\in \mathcal{O}_Z$ $\quad \in \mathcal{O}_X$

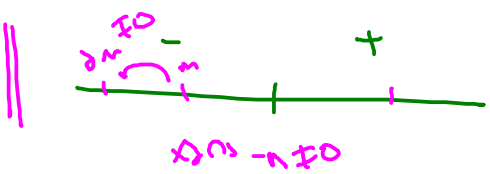
Assume claim:

$i_! = $ "sole" $= \mathrm{coker}(\partial)$

$=$ highest weight part of any "chain"

$i_! = -\otimes k[\partial]$

$\partial x - x\partial = 1$



$D = \partial x = 1 \iff \partial = 0$.

$X = \mathbb{P}^1$

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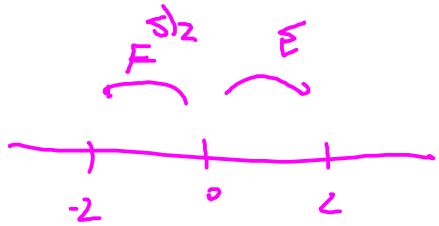

$[G, F] = H$

$2x\partial$